Cheng-Nan Liu

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Summary

8+ years of experience in applying machine learning algorithms, time series analysis, and statistical modeling for large-scale data applications, with 4 peer-reviewed articles in leading geophysical journals.

Education

University of Utah — Salt Lake City, UT
Ph.D. Candidate, Geology and Geophysics — *Aug. 2021 - Aug. 2026 (Expected)*

National Taiwan University — Taipei, Taiwan
M.S., Geoscience — *Aug. 2017 - Jun. 2019*

National Cheng Kung University — Tainan, Taiwan
B.S., Earth Science — *Aug. 2013 - Jun. 2017*

Professional Experience

Data Scientist Intern — Pasadena, CA
Lumetec, Inc. — *Jun. 2025 - Aug. 2025*

Research Scientist Intern — Berkeley, CA
Lawrence Berkeley National Laboratory — *May 2024 - Aug. 2024*

Research Assistant — Taipei, Taiwan
Institute of Earth Sciences, Academia Sinica — *Jan. 2020 – Jul. 2021*

Projects

Vessel Fingerprints System
Lead Developer @ Lumetec.

- Developed a real-time system inspired by the Shazam algorithm to process over 500+ TB of Distributed Acoustic Sensing (DAS) data, enhancing real-time monitoring and protection of undersea fiber-optic cables.
- Developed a system for maritime vessel detection, achieving 70% accuracy for known vessels and 90%+ accuracy for previously undetected vessels, enhancing real-time surveillance capabilities.

Hydrothermal Systems Dynamic Monitoring
Lead Developer @ University of Utah

- Analyzed geophysical, GPS, weather data, and water temperature sensors to uncover drivers of events at Doublet Pool, Old Faithful and Steamboat Geyser in Yellowstone National Park.
- Applied time series and physical modeling to estimate the energy budget and elucidate internalexternal factors.

Time Series Inpainting Generator
Lead Developer @ Lawrence Berkeley National Laboratory

- Led the creation of a Swin Transformer-based algorithm with PyTorch, training over 50 TB of seismic data.
- Speeding up time series reconstruction over 30x faster, from hours to seconds.

Real-time Automated Earthquake Relocation System
Co-developer @ Academia Sinica

- Implementing a Unet-based machine learning model, doubling the number of detected earthquakes with a 10x increase in speed.
- Streamlined and integrated multiple technical modules into a fully automated, real-time system.

Earthquake Hazard Alarm System
Lead Developer @ National Taiwan University

- Designed a multi-branch CNN in Keras, increasing accuracy by 7% and reducing detection time by 0.3 seconds.
- Presented findings at conferences, contributed to publications, and enhanced the Early Warning System in Taiwan.

Skills

Programming Languages: Python, C/C++, MATLAB, Bash, Fortran, SQL
ML & Data Tools: PyTorch, Keras, TensorFlow, NumPy, Pandas, SciPy, Matplotlib, OpenCV, Numba
Technical Skills: Machine Learning, Convolution Neural Network, Transformer Network, Time Series Analysis, Signal Processing, Computer Vision, Statistical Modeling, High-Performance Computing